COFINITE ZEROS OF HIGH DERIVATIVES

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Abstract

We construct a transcendental entire function f such that every nonempty open subset of the complex plane contains a zero of $f^{(n)}$ for all sufficiently large n; equivalently, for every strictly increasing sequence of derivative orders the corresponding zero sets have dense union. The construction is probabilistic. We take a Fock series whose coefficients are independent and uniformly distributed on the closed unit disk, and we combine a saddle-point estimate for the Taylor coefficients of the derivatives with a one-coordinate anti-concentration bound and the mean value property of $\log|f^{(n)}|$ on a zero-free disk. This makes the probability that a derivative is zero-free in a fixed disk summable in n. Every sample of the series obeys the deterministic bound $|f(z)| \leq \sqrt{2}\exp(|z|^2)$, so the function produced has order at most two and finite type.

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1. Introduction

Erdős asked whether there is a nonzero entire function f such that, for every strictly increasing sequence $(n_k)_{k\geq 0}$ of nonnegative integers, the union of the zero sets of $f^{(n_k)}$ is dense in the complex plane [5, p. 72]. The question is recorded as Problem 906 on the Erdős problems page [2]. If polynomials are admitted the question is immediate, because every sufficiently high derivative of a polynomial vanishes identically; we therefore treat the transcendental form of the problem. Throughout, *transcendental entire* means entire and not a polynomial, and $\mathbb{N} = \{0, 1, 2, \ldots\}$. For $c \in \mathbb{C}$ and $R > 0$ we write $B(c, R) = \{z \in \mathbb{C} : |z - c| < R\}$.

For a nonempty open set $U \subset \mathbb{C}$ and an entire function f, define

$$A_U(f) = \{n \in \mathbb{N} : f^{(n)} \text{ has a zero in } U\}.$$

The following elementary reformulation isolates the quantifier that the problem requires.

LEMMA 1.1. *For an entire function f, the following assertions are equivalent.*

(1) *For every strictly increasing sequence* $(n_k)_{k\geq 0}$ *in* $\mathbb{N}$, *the set*

$$\bigcup_{k\geq 0}\{z \in \mathbb{C} : f^{(n_k)}(z) = 0\}$$

is dense in $\mathbb{C}$.

(2) *For every nonempty open set* $U \subset \mathbb{C}$, *the set* $A_U(f)$ *is cofinite in* $\mathbb{N}$, *that is,* $\mathbb{N} \setminus A_U(f)$ *is finite.*

Proof. Assume (2). Fix a strictly increasing sequence (n_k) and a nonempty open set U. Since $A_U(f)$ is cofinite there is an integer $N(U)$ such that $n \geq N(U)$ implies $n \in A_U(f)$. The sequence (n_k) is unbounded, so $n_k \geq N(U)$ for some k, and then $f^{(n_k)}$ has a zero in U. The displayed union therefore meets every nonempty open set and is dense.

Conversely, suppose that (2) fails. Then for some nonempty open set U the set $\mathbb{N} \setminus A_U(f)$ is infinite; enumerate it increasingly as (n_k). Each $f^{(n_k)}$ is zero-free in U, so the union of their zero sets misses U and is not dense. □

Theorem 1.2. *There exists a transcendental entire function* f *such that, for every nonempty open set* $U \subset \mathbb{C}$, *there is an integer* $N(U)$ *for which*

$$n \geq N(U) \quad \Longrightarrow \quad f^{(n)} \text{ has a zero in } U.$$

The same function can be chosen to satisfy

$$|f(z)| \leq \sqrt{2}\exp(|z|^2), \qquad z \in \mathbb{C}. \tag{1}$$

Consequently, the zero sets indexed by every strictly increasing sequence of derivative orders have dense union.

The bound (1) is recorded because it is what the construction produces, and no more is claimed about it here: it says that the function below has order at most two and finite type in the usual terminology of entire-function theory.

Two classical constructions come close to the statement of Theorem 1.2 without reaching it. MacLane [7] built an entire function whose sequence of successive derivatives is dense, in the topology of locally uniform convergence, in the space of all entire functions. Barth and Schneider [1] showed that, for any sequence $(S_k)_{k\geq 0}$ of discrete subsets of $\mathbb{C}$, one can choose derivative orders n_k and a transcendental entire function whose n_k-th derivative vanishes on S_k. Both results yield density statements near the one asked for, in that they force a union of derivative zero sets to be dense; neither yields assertion (2) of Lemma 1.1. MacLane's theorem controls the derivatives only through approximation in the locally uniform topology, which does not preclude an infinite set of derivative orders whose zero sets all miss one fixed disk. In the Barth–Schneider theorem the orders n_k are produced by the construction rather than prescribed in advance, so the conclusion concerns one increasing sequence and not every one. By Lemma 1.1 the problem is exactly the passage from infinitely many admissible derivative orders to cofinitely many, and it is that passage which the construction below supplies.

At the time of access on 5 August 2026, the online record still labelled the problem OPEN and displayed one claimed proof. It also cautions that this label records its owner's current belief and may miss relevant literature [2]. Thus the present article is a proposed resolution, and the status of the question should be assessed from the proof rather than from an online label.

The method is probabilistic and belongs to the theory of random analytic functions; for the general theory of random series of functions we refer to Kahane [6]. The quantity estimated in Section 3, the probability that a random analytic function has no zero in a fixed disk, is a hole probability.

The proof is organised as follows. Section 2 introduces a random Fock series whose coefficients are bounded by one, so that the growth bound (1) holds for every sample, and establishes the two deterministic estimates behind the argument: a saddle-point lower bound identifying a single Taylor coefficient of $f^{(n)}$ that dominates the relevant scale on a given circle, and a matching upper bound for the whole series. Anti-concentration in that one coefficient converts the lower bound into a pointwise tail estimate for the logarithmic deficit of $f^{(n)}$, which is then averaged over a circle. In Section 3, zero-freeness of $f^{(n)}$ on a disk forces, through the mean value property of $\log|f^{(n)}|$, a large averaged deficit on the boundary circle, an event of small probability. The resulting hole probabilities are summable in n; the first Borel–Cantelli lemma [4, Section 2.3] applied along a countable base of disks then produces a single sample with the required property.

2. A bounded random Fock series

Let $\overline{\mathbb{D}}$ be the closed unit disk and let $m_{\mathbb{D}}$ be normalised planar Lebesgue measure on $\overline{\mathbb{D}}$. Put

$$\Omega = \overline{\mathbb{D}}^{\mathbb{N}}, \qquad \mathcal{F} = \mathcal{B}(\overline{\mathbb{D}})^{\otimes\mathbb{N}}, \qquad \mathbb{P} = m_{\mathbb{D}}^{\otimes\mathbb{N}},$$

and equip Ω with the product topology. A countable product of compact metric spaces is compact and metrizable, and $\mathcal{F}$ is the Borel σ-algebra of Ω. For $\omega = (\omega_k)_{k\geq 0} \in \Omega$ set $\xi_k(\omega) = \omega_k$. The coordinate variables $\xi_0, \xi_1, \dots$ are then independent and uniformly distributed on $\overline{\mathbb{D}}$. Define

$$f(z) = \sum_{k=0}^{\infty} \xi_k \frac{z^k}{\sqrt{k!}}. \tag{2}$$

The bound $|\xi_k| \leq 1$ gives normal convergence on compact subsets of $\mathbb{C}$, uniformly in ω. Hence every sample defines an entire function, term by term differentiation is legitimate, and for $n \geq 0$,

$$F_n(z) := f^{(n)}(z) = \sum_{m=0}^{\infty} \xi_{n+m} \frac{\sqrt{(n+m)!}}{m!} z^m. \tag{3}$$

The partial sums of (3) are continuous on $\Omega \times \mathbb{C}$ and converge uniformly on $\Omega \times \{|z| \leq M\}$ for every M, so $(\omega, z) \mapsto F_n(\omega, z)$ is jointly continuous; in particular $\omega \mapsto F_n(\omega, z)$

is $\mathcal{F}$-measurable for each fixed z. The law $m_{\mathbb{D}}$ has no atoms, so $\mathbb{P}\{\xi_k = 0\} = 0$ for every k, and therefore

$$\Omega_0 = \{\omega \in \Omega : \xi_k(\omega) \neq 0 \text{ for every } k \geq 0\} \tag{4}$$

has $\mathbb{P}(\Omega_0) = 1$. On Ω_0 all Taylor coefficients in (2) are nonzero, so f is not a polynomial.

The bounded coefficients also give the growth bound. By Cauchy–Schwarz with weights $2^{k/2}$,

$$\begin{aligned} |f(z)| &\leq \sum_{k\geq 0} \frac{|z|^k}{\sqrt{k!}} \\ &\leq \left(\sum_{k\geq 0} \frac{2^k |z|^{2k}}{k!}\right)^{1/2} \left(\sum_{k\geq 0} 2^{-k}\right)^{1/2} = \sqrt{2}\exp(|z|^2). \end{aligned}$$

Thus (1) holds for every $\omega \in \Omega$, and every sample has order at most two and finite type.

For $r \geq 0$ write

$$a_{n,m}(r) = \frac{\sqrt{(n+m)!}}{m!}\, r^m, \qquad S_n(r) = \sum_{m=0}^{\infty} a_{n,m}(r),$$

and set

$$p_n(z) = \tfrac{1}{2}\log(n!) + |z|\sqrt{n}, \qquad L_n = 2 + \log(n+1). \tag{5}$$

We use the extended-real convention $\log 0 = -\infty$. Every occurrence of L_n below is under hypotheses that force $n \geq 1$.

The saddle-point estimate needs an upper bound for $\log(m!)$. Rather than import a Stirling inequality we prove the following weaker bound, whose loss relative to Stirling's estimate is one logarithm; only the property $L_n = o(\sqrt{n})$ is used, so the loss is harmless.

Lemma 2.1 (Elementary factorial bound). *For every integer $m \geq 1$,*

$$\log(m!) \leq m\log m - m + \log(m+1) + 1.$$

Proof. The function log is nondecreasing on $[1, \infty)$, so $\log j \leq \int_j^{j+1} \log t\, dt$ for every integer $j \geq 1$. Summing over $1 \leq j \leq m$ and evaluating the integral,

$$\log(m!) = \sum_{j=1}^{m} \log j \leq \int_1^{m+1} \log t\, dt = [t\log t - t]_1^{m+1} = (m+1)\log(m+1) - m.$$

Furthermore $(m+1)\log(m+1) = m\log(m+1) + \log(m+1)$, and since $\log(1+u) \leq u$ for $u \geq 0$,

$$m\log(m+1) = m\log m + m\log\Big(1 + \frac{1}{m}\Big) \leq m\log m + 1.$$

Combining the two displays gives the assertion. □

LEMMA 2.2 (Saddle estimates). *Let $n \geq 1$.*

(1) *If $r \geq 0$, $1 \leq r\sqrt{n}$ and $r \leq \sqrt{n}$, then, for $m = \lfloor r\sqrt{n} \rfloor$,*

$$\log a_{n,m}(r) \geq \tfrac{1}{2}\log(n!) + r\sqrt{n} - 2 - \log(n+1);$$

by (5) the right-hand side equals $p_n(z) - L_n$ for every z with $|z| = r$.

(2) *For every $r \geq 0$,*

$$\log S_n(r) \leq \tfrac{1}{2}\log(n!) + r\sqrt{n} + \frac{r + r^2}{2}.$$

Consequently, if $|z| \leq R$ then

$$\log|F_n(z)| - p_n(z) \leq \frac{R + R^2}{2} \tag{6}$$

for every $\omega \in \Omega$, with the inequality read trivially when $F_n(z) = 0$.

PROOF. Factor $a_{n,m}(r) = \sqrt{n!}\, b_{n,m}(r)$, where

$$b_{n,m}(r) = \frac{\sqrt{(n+1)(n+2)\cdots(n+m)}}{m!}\, r^m,$$

which is legitimate because $(n+m)! = n!\,(n+1)(n+2)\cdots(n+m)$.

For (1) put $x = r\sqrt{n}$. Each of the m factors under the square root is at least n, so $b_{n,m}(r) \geq n^{m/2} r^m/m! = x^m/m!$. For $m = \lfloor x \rfloor$ the hypotheses give $1 \leq m \leq x \leq n$, since $x \geq 1$ and $x = r\sqrt{n} \leq \sqrt{n}\sqrt{n} = n$. By Lemma 2.1,

$$\log\frac{x^m}{m!} \geq m\log x - m\log m + m - \log(m+1) - 1 = m\log\frac{x}{m} + m - \log(m+1) - 1.$$

Here $m\log(x/m) \geq 0$ because $m \leq x$; also $m > x - 1$ and $\log(m+1) \leq \log(n+1)$ because $m \leq n$. Hence

$$\log\frac{x^m}{m!} \geq (x-1) - \log(n+1) - 1 = x - 2 - \log(n+1),$$

and adding $\frac{1}{2}\log(n!)$ to both sides gives (1); the identification of the right-hand side with $p_n(z) - L_n$ is the definition (5).

For (2), note that

$$b_{n,m}(r) = \sqrt{\frac{\binom{n+m}{n}}{m!}}\, r^m.$$

If $r > 0$, put $t = r/(r + \sqrt{n}) \in (0,1)$, so that $\binom{n+m}{n} r^{2m}/m! = (\binom{n+m}{n} t^m)((r^2/t)^m/m!)$. Cauchy–Schwarz, the binomial series $\sum_{m\geq 0}\binom{n+m}{n} t^m = (1-t)^{-(n+1)}$ and the exponential series give

$$\begin{aligned}\sum_{m\geq 0} b_{n,m}(r) &\leq \left(\sum_{m\geq 0}\binom{n+m}{n} t^m\right)^{1/2}\left(\sum_{m\geq 0}\frac{(r^2/t)^m}{m!}\right)^{1/2}\\ &= (1-t)^{-(n+1)/2}\exp\left(\frac{r^2}{2t}\right).\end{aligned}$$

Put $u = r/\sqrt{n}$. Since $\log(1+u) \le u$ for $u \ge 0$ and $\sqrt{n} \ge 1$,

$$\begin{aligned}\frac{n+1}{2}\log(1+u) + \frac{r(r+\sqrt{n})}{2} &\le \frac{(n+1)r}{2\sqrt{n}} + \frac{r^2 + r\sqrt{n}}{2} \\ &= r\sqrt{n} + \frac{r}{2\sqrt{n}} + \frac{r^2}{2} \\ &\le r\sqrt{n} + \frac{r+r^2}{2}.\end{aligned}$$

Adding $\frac{1}{2}\log(n!)$ gives (2) for $r > 0$; the case $r = 0$ is immediate because $S_n(0) = \sqrt{n!}$. Finally $|\xi_{n+m}| \le 1$, so (3) and the triangle inequality give $|F_n(z)| \le S_n(|z|)$. Applying the bound just proved with $r = |z|$, and using that $r \mapsto (r + r^2)/2$ is nondecreasing, gives (6) whenever $|z| \le R$. □

The second ingredient is anti-concentration. If ξ is uniform on $\overline{\mathbb{D}}$ then, for $a \ne 0$, $w \in \mathbb{C}$ and $\varepsilon \ge 0$,

$$\mathbb{P}\{|a\xi + w| < \varepsilon\} \le \left(\frac{\varepsilon}{|a|}\right)^2, \tag{7}$$

because the set $\{\zeta : |a\zeta + w| < \varepsilon\}$ is the disk of centre $-w/a$ and radius $\varepsilon/|a|$, whose normalised area is at most $(\varepsilon/|a|)^2$.

Lemma 2.3 (Pointwise logarithmic tail). *Let $n \ge 1$ and let $z \in \mathbb{C}$ satisfy $1 \le |z|\sqrt{n}$ and $|z| \le \sqrt{n}$. Then, for every $s \ge 0$,*

$$\mathbb{P}\{p_n(z) - \log|F_n(z)| > s + L_n\} \le e^{-2s}. \tag{8}$$

Proof. Put $m = \lfloor |z|\sqrt{n} \rfloor$ and $k = n + m$, and split (3) by isolating the coefficient ξ_k:

$$F_n(\omega, z) = a\,\xi_k(\omega) + w(\omega),$$

where

$$a = \frac{\sqrt{(n+m)!}}{m!}\,z^m$$

is deterministic and

$$w(\omega) = \sum_{j \ne m} \xi_{n+j}(\omega)\frac{\sqrt{(n+j)!}}{j!}\,z^j$$

is the remainder. Three properties of w are used. First, the series defining w converges absolutely for every ω, since $|\xi_{n+j}| \le 1$ and $\sum_{j\ge 0} a_{n,j}(|z|) = S_n(|z|) < \infty$ by Lemma 2.2(2); so w is finite everywhere, not merely almost surely. Second, w is a measurable function of ω, being a uniform limit of continuous functions on Ω. Third, and decisively, w does not depend on the coordinate ω_k: it is measurable with respect to $\mathcal{G} = \sigma(\xi_j : j \ne k)$, since the index $j = m$ is omitted from the sum. Note also that $|z| \ge 1/\sqrt{n} > 0$, so $a \ne 0$ and $|a| = a_{n,m}(|z|)$.

Since $|z| =: r$ satisfies the hypotheses of Lemma 2.2(1), that lemma gives $\log|a| \geq p_n(z) - L_n$. Hence, for $s \geq 0$,

$$\{p_n(z) - \log|F_n(z)| > s + L_n\} = \{|F_n(z)| < e^{\,p_n(z)-L_n-s}\} \subseteq \{|a\xi_k + w| < |a|e^{-s}\}.$$

Now write $\Omega = \overline{\mathbb{D}} \times \Omega'$, where the first factor carries the coordinate ω_k and $\Omega' = \overline{\mathbb{D}}^{\mathbb{N}\setminus\{k\}}$ carries the remaining coordinates, and correspondingly $\mathbb{P} = m_{\mathbb{D}} \otimes \mathbb{P}'$. Because w is $\mathcal{G}$-measurable it is a function of $\omega' \in \Omega'$ alone, so Tonelli's theorem gives

$$\mathbb{P}\{|a\xi_k + w| < |a|e^{-s}\} = \int_{\Omega'} m_{\mathbb{D}}\{\zeta \in \overline{\mathbb{D}} : |a\zeta + w(\omega')| < |a|e^{-s}\}\, d\mathbb{P}'(\omega').$$

For each fixed ω' the inner measure is at most $(e^{-s})^2 = e^{-2s}$ by (7), with $\varepsilon = |a|e^{-s}$; the bound is uniform in ω', so integrating over Ω' preserves it. This proves (8). □

The pointwise estimate is used through the following circle-average bound.

Lemma 2.4 (Circle lower tail). *Let $Y \geq 0$ be jointly measurable on $\Omega \times [0, 2\pi]$ and suppose that*

$$\mathbb{P}\{Y(\cdot, \theta) > s\} \leq e^{-s/2}, \qquad s \geq 0,$$

uniformly in θ. Put $\overline{Y} = (2\pi)^{-1} \int_0^{2\pi} Y(\cdot, \theta)\, d\theta$. Then

$$\mathbb{P}\{\overline{Y} \geq t\} \leq 2e^{-t/8}, \qquad t \geq 8.$$

Proof. The tail hypothesis and the layer-cake formula give $\mathbb{E}Y(\cdot, \theta) = \int_0^\infty \mathbb{P}\{Y(\cdot, \theta) > s\}\, ds \leq 2$ for every θ, so $\mathbb{E}\overline{Y} \leq 2$ by Tonelli's theorem and $\overline{Y}$ is integrable. Put $E = \{\overline{Y} \geq t\}$ and $q = \mathbb{P}(E)$. Then $t\mathbf{1}_E \leq \overline{Y}\mathbf{1}_E$, and Tonelli's theorem gives

$$tq \leq \mathbb{E}(\overline{Y}\mathbf{1}_E) = \frac{1}{2\pi}\int_0^{2\pi} \mathbb{E}(Y(\cdot, \theta)\mathbf{1}_E)\, d\theta.$$

For every θ the layer-cake formula and $\mathbb{P}(\{Y(\cdot, \theta) > s\} \cap E) \leq \min\{q, e^{-s/2}\}$ yield

$$\mathbb{E}(Y(\cdot, \theta)\mathbf{1}_E) \leq \int_0^\infty \min\{q, e^{-s/2}\}\, ds = \int_0^{2\log(1/q)} q\, ds + \int_{2\log(1/q)}^\infty e^{-s/2}\, ds = 2q \log\frac{e}{q},$$

where the computation assumes $0 < q \leq 1$. If $q > 0$ we may cancel q and obtain $t \leq 2\log(e/q) = 2 + 2\log(1/q)$, so $q \leq e^{1-t/2}$. For $t \geq 8$ we have $e^{1-t/2} = e^{1-3t/8}e^{-t/8} \leq e^{-2}e^{-t/8} \leq 2e^{-t/8}$. The case $q = 0$ is immediate. □

3. Zero-free disks and proof of the theorem

We first isolate the geometric gain that drives the estimate. For $c \in \mathbb{C}$ and $r > 0$ set

$$\gamma(c, r) = \frac{1}{2\pi}\int_0^{2\pi} |c + re^{i\theta}|\, d\theta - |c|.$$

Lemma 3.1. *For all $c \in \mathbb{C}$ and $r > 0$ we have $\gamma(c, r) > 0$.*

Proof. Define

$$\varphi(\theta) = |c + re^{i\theta}| + |c - re^{i\theta}| - 2|c|, \qquad \theta \in [0, 2\pi].$$

Since $(c + re^{i\theta}) + (c - re^{i\theta}) = 2c$, the triangle inequality gives $\varphi \geq 0$ everywhere, and φ is continuous because $\theta \mapsto |c \pm re^{i\theta}|$ is.

We claim that $\varphi(\theta_0) > 0$ for some θ_0. If $c = 0$ then $\varphi \equiv 2r > 0$. If $c \neq 0$, choose θ_0 with $re^{i\theta_0} = irc/|c|$, so that $re^{i\theta_0}$ is perpendicular to c; then $|c \pm re^{i\theta_0}| = \sqrt{|c|^2 + r^2}$ by the Pythagorean identity, so

$$\varphi(\theta_0) = 2\sqrt{|c|^2 + r^2} - 2|c| > 0.$$

This is the point at which continuity is needed: perpendicularity produces strict inequality only at the two isolated angles θ_0 and $\theta_0 + \pi$, and a set of measure zero contributes nothing to an integral. Since φ is continuous and $\varphi(\theta_0) > 0$, there is a $\delta > 0$ such that

$$\varphi(\theta) \geq \tfrac{1}{2}\varphi(\theta_0) > 0 \qquad \text{for all } \theta \in I := [\theta_0 - \delta, \theta_0 + \delta], \tag{9}$$

so strictness holds on an interval of positive length, not merely at isolated angles.

The substitution $\theta \mapsto \theta + \pi$ and 2π-periodicity give $\int_0^{2\pi} |c - re^{i\theta}|\, d\theta = \int_0^{2\pi} |c + re^{i\theta}|\, d\theta$, whence

$$\frac{1}{2\pi}\int_0^{2\pi} \varphi(\theta)\, d\theta = 2\left(\frac{1}{2\pi}\int_0^{2\pi} |c + re^{i\theta}|\, d\theta - |c|\right) = 2\gamma(c, r).$$

Since $\varphi \geq 0$ everywhere, (9) gives

$$2\gamma(c, r) \geq \frac{1}{2\pi}\int_I \varphi(\theta)\, d\theta \geq \frac{2\delta}{2\pi} \cdot \tfrac{1}{2}\varphi(\theta_0) > 0,$$

which is the assertion. (Here I is understood modulo 2π; if it is not contained in $[0, 2\pi]$, replace it by its intersection with a period interval containing θ_0 in its interior.) □

Next we record the mean value step in the exact form needed, with the strict radius margin made explicit.

Lemma 3.2 (Mean value property on a zero-free disk). *Let $c \in \mathbb{C}$ and $\rho > 0$, let G be holomorphic and zero-free on $B(c, \rho)$, and let* $0 < r < \rho$. *Then*

$$\frac{1}{2\pi}\int_0^{2\pi} \log|G(c + re^{i\theta})|\, d\theta = \log|G(c)|.$$

Proof. The disk $B(c, \rho)$ is convex, and G'/G is holomorphic on it because G is zero-free there. Every holomorphic function on a convex domain has a holomorphic primitive [3, Chapter IV], so there is a holomorphic h on $B(c, \rho)$ with $h' = G'/G$. Then $(Ge^{-h})' = e^{-h}(G' - Gh') = 0$ on the connected set $B(c, \rho)$, so $Ge^{-h} \equiv \kappa$ for a constant κ, and $\kappa \neq 0$ because G is zero-free. Choose $\lambda \in \mathbb{C}$ with $e^{\lambda} = \kappa$ and set $g = h + \lambda$; then $e^g = G$ on $B(c, \rho)$, and hence $\operatorname{Re} g = \log|G|$ there.

The strict inequality $r < \rho$ gives $\overline{B(c,r)} \subset B(c,\rho)$, so the circle $|z-c| = r$ and its interior lie in the domain of g and Cauchy's integral formula [3, Chapter IV] applies:

$$g(c) = \frac{1}{2\pi i}\int_{|z-c|=r} \frac{g(z)}{z-c}\,dz = \frac{1}{2\pi}\int_0^{2\pi} g(c+re^{i\theta})\,d\theta,$$

the second equality by the parametrisation $z = c + re^{i\theta}$, $dz = ire^{i\theta}\,d\theta$. Taking real parts gives the assertion. □

Now fix $c \in \mathbb{Q} + i\mathbb{Q}$ with $c \neq 0$ and a rational R with $0 < R < |c|$, and put

$$r = \frac{R}{2} < R, \tag{10}$$

so that $\overline{B(c,r)} \subset B(c,R)$ with a strict radius margin. On the circle $|z-c| = r$ we have

$$0 < \delta := |c| - r \leq |z| \leq |c| + r =: M,$$

the left inequality because $r < R < |c|$. Consequently there is an integer $n_0 = n_0(c,R) \geq 1$ such that

$$1 \leq \delta\sqrt{n} \quad \text{and} \quad M \leq \sqrt{n} \qquad \text{for all } n \geq n_0, \tag{11}$$

and then the hypotheses of Lemma 2.3 hold at every point of the circle $|z-c| = r$, uniformly in θ.

To keep the logarithmic deficit real-valued, define

$$\ell(w) = \begin{cases} \log|w|, & w \neq 0, \\ 0, & w = 0, \end{cases} \qquad D_{n,z} = \max\{p_n(z) - \ell(F_n(z)) - L_n,\ 0\},$$

and, for $n \geq n_0$,

$$\overline{D}_n = \frac{1}{2\pi}\int_0^{2\pi} D_{n,c+re^{i\theta}}\,d\theta.$$

The function ℓ is Borel measurable on $\mathbb{C}$, and $(\omega, z) \mapsto F_n(\omega, z)$ is jointly continuous, so $(\omega,\theta) \mapsto D_{n,c+re^{i\theta}}(\omega)$ is jointly measurable on $\Omega \times [0, 2\pi]$ and $\overline{D}_n$ is a measurable function of ω. Because $\ell(w) \geq \log|w|$ for every $w \in \mathbb{C}$ in the extended-real order, with equality unless $w = 0$, we have, for $s \geq 0$,

$$\{D_{n,z} > s\} \subseteq \{p_n(z) - \log|F_n(z)| > s + L_n\}.$$

Combining this with Lemma 2.3 and using $e^{-2s} \leq e^{-s/2}$ for $s \geq 0$ gives the tail hypothesis of Lemma 2.4 explicitly: for $n \geq n_0$, every z on the circle $|z-c| = r$ and every $s \geq 0$,

$$\mathbb{P}\{D_{n,z} > s\} \leq \mathbb{P}\{p_n(z) - \log|F_n(z)| > s + L_n\} \leq e^{-2s} \leq e^{-s/2}. \tag{12}$$

So Lemma 2.4 applies to $Y(\omega,\theta) = D_{n,c+re^{i\theta}}(\omega)$ and to $\overline{Y} = \overline{D}_n$.

Let

$$H_n(c,R) = \{\omega \in \Omega : F_n(\omega,\cdot) \text{ has no zero in } B(c,R)\}.$$

LEMMA 3.3. $H_n(c,R) \in \mathcal{F}$.

PROOF. We claim that

$$H_n(c,R) = \bigcap_{\substack{\rho\in\mathbb{Q} \\ 0<\rho<R}} \Big\{\omega : \min_{|z-c|\le\rho} |F_n(\omega,z)| > 0\Big\}, \tag{13}$$

the minimum being attained because $z \mapsto |F_n(\omega,z)|$ is continuous and $\overline{B(c,\rho)}$ is compact. If $\omega \in H_n(c,R)$ and $0 < \rho < R$, then $\overline{B(c,\rho)} \subset B(c,R)$, so $|F_n(\omega,\cdot)|$ is continuous and strictly positive on a compact set and its minimum there is positive. Conversely, if the right-hand side contains ω and $z \in B(c,R)$, choose a rational ρ with $|z-c| \le \rho < R$; then $|F_n(\omega,z)| > 0$. This proves (13), and the intersection is countable.

It remains to check that, for fixed $\rho \in (0,R)$, the function

$$M_\rho(\omega) = \min_{|z-c|\le\rho} |F_n(\omega,z)|$$

is $\mathcal{F}$-measurable. Let $Q_\rho = (\mathbb{Q}+i\mathbb{Q}) \cap B(c,\rho)$, a countable set that is dense in $\overline{B(c,\rho)}$. For fixed ω the function $z \mapsto |F_n(\omega,z)|$ is continuous on the compact set $\overline{B(c,\rho)}$, and a continuous function on a compact set has the same infimum over the set as over any dense subset of it; hence

$$M_\rho(\omega) = \inf_{z\in Q_\rho} |F_n(\omega,z)|.$$

For each fixed z the map $\omega \mapsto |F_n(\omega,z)|$ is measurable, as noted after (3), so M_ρ is a countable infimum of measurable functions and is therefore measurable. Consequently $\{M_\rho > 0\} \in \mathcal{F}$ for every rational $\rho \in (0,R)$, and (13) exhibits $H_n(c,R)$ as a countable intersection of sets in $\mathcal{F}$. □

We can now estimate $\mathbb{P}(H_n(c,R))$. Fix $n \ge n_0$ and work on the event $H_n(c,R)$, on which F_n is holomorphic and zero-free on $B(c,R)$ by the definition of that event. Since $r = R/2 < R$ by (10), Lemma 3.2 applies with $G = F_n$ and $\rho = R$ and gives

$$\frac{1}{2\pi}\int_0^{2\pi} \log|F_n(c+re^{i\theta})|\,d\theta = \log|F_n(c)|. \tag{14}$$

Lemma 2.2(2), applied with R replaced by $|c|$ and evaluated at $z = c$, gives

$$\log|F_n(c)| \le p_n(c) + C(c), \qquad C(c) = \frac{|c|+|c|^2}{2}. \tag{15}$$

On $H_n(c,R)$ we have $F_n(z) \ne 0$ for $z \in \overline{B(c,r)}$, so $\ell(F_n(z)) = \log|F_n(z)|$ there and therefore $D_{n,z} \ge p_n(z) - \log|F_n(z)| - L_n$ on the circle $|z-c| = r$. Averaging over that circle and using

$$\frac{1}{2\pi}\int_0^{2\pi} p_n(c+re^{i\theta})\,d\theta = \tfrac{1}{2}\log(n!) + \sqrt{n}\cdot\frac{1}{2\pi}\int_0^{2\pi} |c+re^{i\theta}|\,d\theta = p_n(c) + \gamma(c,r)\sqrt{n},$$

together with (14) and (15), we obtain on $H_n(c,R)$ that

$$\overline{D}_n \geq p_n(c) + \gamma(c,r)\sqrt{n} - \log|F_n(c)| - L_n \geq \gamma(c,r)\sqrt{n} - C(c) - L_n =: T_n,$$

that is,

$$H_n(c,R) \subseteq \{\overline{D}_n \geq T_n\}, \qquad T_n = \gamma(c,r)\sqrt{n} - C(c) - 2 - \log(n+1). \tag{16}$$

By Lemma 3.1 we have $\gamma(c,r) > 0$, and $C(c) + 2 + \log(n+1) = o(\sqrt{n})$, so there is $n_1 = n_1(c,R) \geq n_0$ with

$$T_n \geq \max\Big\{\tfrac{1}{2}\gamma(c,r)\sqrt{n},\ 8\Big\}, \qquad n \geq n_1. \tag{17}$$

For $n \geq n_1$, Lemma 2.4 together with (12), (16) and (17) gives

$$\mathbb{P}(H_n(c,R)) \leq \mathbb{P}\{\overline{D}_n \geq T_n\} \leq 2e^{-T_n/8} \leq 2\exp\left(-\frac{\gamma(c,r)}{16}\sqrt{n}\right). \tag{18}$$

The right-hand side of (18) is summable, since $\exp(-\gamma(c,r)\sqrt{n}/16) \leq n^{-2}$ for all large n, and the finitely many terms with $n < n_1$ are each at most 1. Hence

$$\sum_{n=0}^{\infty} \mathbb{P}(H_n(c,R)) < \infty. \tag{19}$$

All the events involved are measurable by Lemma 3.3, so the first Borel–Cantelli lemma [4, Section 2.3] applies and shows that

$$\mathbb{P}\Big(\limsup_{n\to\infty} H_n(c,R)\Big) = 0; \tag{20}$$

that is, almost surely only finitely many F_n are zero-free in $B(c,R)$.

Proof of Theorem 1.2. Let $\mathcal{B}$ be the family of disks $B(c,R)$ with $c \in \mathbb{Q} + i\mathbb{Q}$, $c \neq 0$, $R \in \mathbb{Q}_{>0}$ and $R < |c|$; it is countable. We check that every nonempty open set $U \subset \mathbb{C}$ contains a member of $\mathcal{B}$. Since U is open and nonempty it contains a point $z_0 \neq 0$, and there is $\varepsilon > 0$ with $B(z_0,\varepsilon) \subset U$ and $\varepsilon < |z_0|/2$. Choose $c \in \mathbb{Q} + i\mathbb{Q}$ with $|c - z_0| < \varepsilon/4$; then $|c| \geq |z_0| - \varepsilon/4 > 0$, so $c \neq 0$. Choose a rational R with $0 < R < \min\{\varepsilon/2, |c|\}$. Then $R < |c|$, and every z with $|z - c| < R$ satisfies $|z - z_0| < \varepsilon/4 + \varepsilon/2 < \varepsilon$, so $B(c,R) \subset B(z_0,\varepsilon) \subset U$ and $B(c,R) \in \mathcal{B}$.

By (20), for each $B = B(c,R) \in \mathcal{B}$ there is a $\mathbb{P}$-null set $\mathcal{N}_B$ outside of which only finitely many F_n are zero-free in B. The union $\mathcal{N} = \bigcup_{B\in\mathcal{B}} \mathcal{N}_B$ is a countable union of null sets and is therefore null. With Ω_0 as in (4) we have $\mathbb{P}(\Omega_0 \setminus \mathcal{N}) = 1$; in particular this event is nonempty. Fix $\omega \in \Omega_0 \setminus \mathcal{N}$ and let f be the entire function (2) determined by it.

Then f is transcendental, because $\omega \in \Omega_0$ makes every Taylor coefficient of f nonzero, and f satisfies (1), which holds for every sample. Let $U \subset \mathbb{C}$ be nonempty and open and choose $B \in \mathcal{B}$ with $B \subset U$. Since $\omega \notin \mathcal{N}_B$, only finitely many $F_n = f^{(n)}$ are zero-free in B. Let $N(U)$ be one more than the largest such index, or let $N(U) = 0$ if there is no such index. Then every $n \geq N(U)$ gives a zero of $f^{(n)}$ in B, hence in U. This is the first assertion of Theorem 1.2, and the last assertion follows from it by Lemma 1.1. □

Declarations

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